

KDX1/Mlp1 (P36005, YKL161C) — Adjudication of Cell-Cycle, CDK-Complex, and ATP-Binding Function Assignments

Focus type: function_assignment · **Hypothesis slug:** cell-cycle-complex-and-nucleotide-binding-functions **Organism:** *Saccharomyces cerevisiae* (NCBITaxon:559292) · **Gene:** KDX1 (synonym MLP1) · **ORF:** YKL161C

Executive Judgment

Verdict: Over-annotated / Refuted for the cell-cycle & CDK-complex claims; Weakly-supported→Unresolved for ATP binding.

The four seed candidate functions are adjudicated independently:

Candidate function	Current GO (evidence)	Verdict
Regulates cell cycle / G1/S–G2/M transition	GO:0000086 G2/M transition (IBA)	Refuted as a direct function / over-annotated
Cyclin-dependent protein kinase holoenzyme complex	GO:0000307 (IBA)	Refuted / over-annotated
Cyclin-dependent protein Ser/Thr kinase activity	GO:0004693 (IBA)	Refuted / over-annotated
Binds ATP	GO:0005524 (IEA:InterPro)	Weakly supported → Unresolved (rule-based only; likely impaired)

Reasoning. KDX1/Mlp1 is a **pseudokinase** — a whole-genome-duplication paralog of the active cell-wall-integrity (CWI) MAPK Slt2/Mpk1. Its only experimentally established molecular function is **noncatalytic transcriptional coactivation**: together with (and redundantly to) Mpk1, it forms a complex with the Swi4/Swi6 (SBF) transcription factor to drive cell-wall-stress genes (FKS2 and others) independently of kinase activity

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It is a MAPK-family (CMGC group) protein, **not** a cyclin-dependent kinase. The CDK-activity, CDK-holoenzyme-complex, and G2/M-transition annotations are **IBA (PANTHER/PAINT) propagations** from a broad CMGC-group ancestor that co-mingles CDK and MAPK subfamilies; they are not supported by any KDX1 primary experiment and conflict with the protein's demonstrated biology. ATP binding is an **IEA:InterPro rule-based** call from the generic protein-kinase profile; sequence analysis shows KDX1 has lost the DFG Mg²⁺-coordinating aspartate (→NFG) and has a degraded glycine-rich loop, so Mg-ATP binding is likely impaired and remains experimentally undemonstrated.

Most important caveats. (1) Absence of a direct ATP-binding assay means we cannot formally exclude residual nucleotide binding (the β 3 catalytic Lys55 is retained). (2) SBF (Swi4/Swi6) does have a bona-fide G1/S cell-cycle role, so a *tenuous, indirect* cell-cycle connection exists via its partner — but this is a downstream/partner property, not a direct KDX1 function. (3) The seed's caution is correct: a NOT annotation from a different fungal PANTHER node (PTN008603245) should not be transferred to this target without established ancestry.

Evidence Matrix

Citation (PMID)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Con- lim
18268013 (Kim, Truman, Levin 2008)	Mutant/ biochemical (ChIP, kinase-dead)	Refutes catalytic role; supports noncatalytic SBF function	Does Mlp1/Mpk1 act via kinase activity or noncatalytically?	Mpk1 and its pseudokinase paralog Mlp1 activate FKS2 via a noncatalytic mechanism; form dimeric complex with Swi4, recruit Swi6; FKS2 activation independent of protein kinase activity	<i>S. cerevisiae</i> , cell-wall stress, FKS2 promoter	Hig me Dir Mlp pse
20641022 (Kim & Levin 2010)	Reporter/ genetic	Supports; Qualifies redundancy	Is Mlp1 role redundant with Mpk1? Which genes?	"The Mlp1 pseudokinase serves a redundant function with Mpk1"; additional SBF-noncatalytic targets CHA1, YLR042c, YKR013w	<i>S. cerevisiae</i> CWI transcription	Hig Con red non SBF dep role
20219973 (Kim et al 2010)	Biochemical/ localization	Qualifies; contextualizes SBF	Mechanism of Swi6 recruitment	Mpk1(Mlp1)–Swi4 complex recruits Swi6 to nucleus for CWI transcription; Mpk1 phosphorylates Swi6 Ser238	<i>S. cerevisiae</i>	Hig Mpk sha con form step (no

Citation (PMID)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Con- lim
20489023 (Breitkreutz et al 2010, <i>Science</i>)	Interaction (systematic network)	Competing/ context (not CDK evidence)	Seed's 3rd cited PMID — what is it?	Global yeast kinase–phosphatase interaction network (large-scale AP-MS); a systematic interactome resource, not a functional CDK-complex study	<i>S. cerevisiae</i> , proteome-scale	Ver- run- eSu- Exp- syst- inte- dat- dat- doe- KDX- CDK- hol-
UniProt P36005 (features)	Computational/ rule-based	Qualifies ATP-binding claim	Are ATP-binding sites experimental?	P-loop (29–37), Lys55, active-site 153 all annotated by PROSITE-ProRule (ECO:0000255) ; GO:0005524 = IEA:InterPro	Sequence rule	Rul- only assa- con- bin-
UniProt GO xrefs	Database (IBA/ IEA)	Competing/ context	Provenance of CDK/cell-cycle terms	GO:0000307, GO:0004693, GO:0000086 are all IBA:GO_Central (PAINT); no experimental support	PANTHER family tree	The- term adj- pro- phy- infe- exp-
This report — sequence analysis (P36005 vs P32485)	Structural/ evolutionary (computed)	Refutes catalytic competence; qualifies ATP binding	Does KDX1 retain kinase catalytic machinery?	KDX1 vs Slr2: DFG→NFG (Mg-Asp lost), HRD→HCD (Arg→Cys),	In-silico, public sequences	Hig- call- bin- con-

Citation (PMID)	Evidence type	Supports/Refutes/Qualifies	Claim tested	Key finding	Context	Con lim
				degraded P-loop (no GxGxxG), no MAPK T-x-Y activation motif ; β 3 Lys retained		is in not
This report — InterPro/PANTHER/CDD (P36005)	Computational/database	Refutes CDK classification; competing with CDK IBA terms	Which kinase family does KDX1 belong to?	PANTHER PTHR24055 "MAPK" ; InterPro family IPR050117 "Mitogen-activated protein kinase" ; CDD cd07857 = fungal MPK1 (SlT2) MAPK clade . ATP-binding GO derives from IPR017441 / PROSITE PS00107 ATP-binding <i>signature</i> (motif pattern), no CDK family present	InterPro protein2ipr	Hig Cor MA CDK term cro ove pro

GO Curation Implications (leads — require curator verification)

- **GO:0000307 cyclin-dependent protein kinase holoenzyme complex (CC, IBA):** *Lead — REMOVE / do-not-assert.* No evidence KDX1 is in a CDK–cyclin complex; it is a MAPK-family pseudokinase whose documented complex is with Swi4/Swi6 (SBF). Consider replacing with a supported CC such as the transcription-factor/SBF context (or leaving CC to the nucleus based on its SBF role).
 - **GO:0004693 CDK Ser/Thr kinase activity (MF, IBA):** *Lead — REMOVE / do-not-assert.* Pseudokinase; no catalytic activity, and even the parent kinase-activity call is questionable. If any MF is kept it should reflect the noncatalytic scaffolding role, not CDK activity.
 - **GO:0004674 / GO:0106310 protein (Ser/Thr) kinase activity (ISS/IEA):** *Lead — flag for review.* Contradicted by pseudokinase status (DFG→NFG, no activation motif); at minimum should not be a core function; a NOT-qualifier is defensible if curator confirms no residual activity.
 - **GO:0000086 G2/M transition of mitotic cell cycle (BP, IBA):** *Lead — REMOVE / do-not-assert as direct function.* No KDX1 experiment supports G2/M control; any cell-cycle link is indirect through its SBF partner (which is G1/S, not G2/M).
 - **GO:0005524 ATP binding (MF, IEA:InterPro):** *Lead — treat as uncertain.* Rule-based only; sequence predicts impaired Mg-ATP binding. Do not upgrade to experimental; acceptable to retain as low-confidence IEA or flag pending assay.
 - **Supported terms to retain/strengthen (positive leads):** BP cell-wall-stress transcriptional response / **GO:0071554 cell wall organization or biogenesis** context; MF/BP capturing **noncatalytic transcription coactivator activity** with SBF (e.g., a transcription-coregulator MF is more informative than "protein binding"); CC **nucleus** (promoter-bound complex). Interaction with Rlm1 and Swi4/Swi6 supports specific complex/coregulator terms rather than generic "protein binding."
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Mechanistic Scope

Immediate molecular function tested: whether KDX1 directly (a) has CDK/kinase catalytic activity, (b) belongs to a CDK holoenzyme complex, (c) directly controls cell-cycle transitions, and (d) binds ATP.

- **Direct gene-product activity that is supported:** noncatalytic assembly with Swi4 (and Swi6) to confer DNA-binding competence and activate cell-wall-stress transcription — redundant with Mpk1.
 - **Downstream / partner-derived, not direct:** any cell-cycle connotation arises from SBF's separate G1/S role, not from KDX1 acting on the cell-cycle machinery. G2/M is not mechanistically linked at all.
 - **Inferred, not assayed:** ATP/nucleotide binding — predicted impaired from loss of DFG-Asp and P-loop degradation; no phosphotransfer expected (no activation motif; HRD-Arg lost).
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Conflicts and Alternatives

- **Paralog/namesake confusion (flagged in seed, confirmed relevant):** "MLP1" is also the name of the *nuclear-pore / myosin-like protein Mlp1* (YKR095W, UniProt Q02455), a ~1875-aa nuclear-basket coiled-coil protein, and its paralog **Mlp2** (YIL149C, P40457, 1679 aa) which "associate with the nuclear pore complex" and organise perinuclear chromatin (verified via UniProt this run). The target here is **KDX1/Mlp1 = YKL161C, P36005** (433 aa, Slt2 pseudokinase) — a completely different, much smaller protein. Curators must not import nuclear-basket / NPC / mRNA-export annotations onto P36005.
- **PANTHER/PAINT over-propagation (confirmed):** InterPro/PANTHER/CDD place P36005 unambiguously in the **MAPK family (PTHR24055; IPR050117)** and specifically the **fungus MPK1/Slt2 MAPK clade (CDD cd07857)** — no CDK-family assignment exists. The CDK-activity, CDK-holoenzyme, and G2/M IBA terms therefore cannot come from KDX1's own subfamily and must be propagated from a broad CMGC-group ancestor node that mixes CDK and MAPK clades. This is a classic family-tree over-annotation; the MAPK subfamily should not inherit CDK-specific MF/CC/BP terms. The ATP-binding IEA traces to the **PROSITE PS00107 / IPR017441 ATP-binding signature** (a sequence motif pattern), not to any binding assay.

- **NOT-annotation transfer risk:** The seed correctly warns that a NOT at another fungal node (PTN008603245) must not be applied unless ancestry to leaf PTN000623199 is established. My recommendation to *not assert* CDK/cell-cycle terms does not depend on importing that NOT; it rests on the target's own biology and sequence.
- **Genuine but indirect cell-cycle thread:** Because SBF (Swi4/Swi6) governs G1/S, and KDX1 physically joins SBF under stress, a curator could see spurious support for "cell-cycle." This is an alternative interpretation to guard against, not evidence for a direct KDX1 cell-cycle function.

Knowledge Gaps

1. **Direct ATP/nucleotide binding of KDX1.** Checked: sequence (DFG→NFG, degraded P-loop, retained Lys55) and UniProt provenance (rule-based). Matters because GO:0005524 rests entirely on an InterPro rule. Resolve with: ITC/DSF thermal-shift or ADP-Glo nucleotide-binding assay on recombinant Kdx1; or a structure/AlphaFold-nucleotide-pocket analysis.
2. **Any residual catalytic activity.** Checked: motif loss predicts none. Matters for whether a NOT-kinase-activity qualifier is warranted. Resolve with: in-vitro kinase assay (already implied negative by kinase-dead-independent transcription in

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but not directly assayed for Kdx1 alone).

3. **PANTHER tree ancestry for the disputed IBA terms.** Checked: UniProt lists them as IBA:GO_Central. Matters because the correct curation is to prune CDK terms from the MAPK leaf. Resolve with: inspection of PAINT nodes PTN001172579/PTN000623979/PTN000623980 vs leaf PTN000623199 to confirm the terms derive from CDK-clade ancestors.
 4. **Whether the SBF-linked role justifies any cell-cycle BP term.** Checked: literature ties Mlp1 to stress transcription, not cell-cycle progression. Resolve with: cell-cycle phenotyping of *kdx1Δ* (synchronised cultures) — expected null given redundancy and stress-specific role.
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Discriminating Tests

- **Nucleotide-binding assay** (DSF/thermal shift $\pm$ ATP-Mg²⁺; ADP-Glo) on recombinant Kdx1 vs Slt2 — cleanly separates "ATP binding" from "no binding."
- **Structure-based pocket check:** AlphaFold model of P36005 $\rightarrow$ measure ATP-pocket integrity (DFG-Asp absent, P-loop geometry) and compare to Slt2 — supports/undermines GO:0005524 conservatively.
- **Cell-cycle assay of kdx1 Δ and slt2 Δ kdx1 Δ** (FACS/bud-index, synchronised) — tests any real G1/S or G2/M contribution.
- **PANTHER PAINT node audit** (PTN000623199 leaf and ancestors) — confirms the IBA CDK/ cell-cycle terms originate from a CDK-clade ancestor and should be pruned for the MAPK leaf.
- **ChIP/co-IP of Kdx1 with Swi4/Swi6 and Rlm1** — strengthens the correct complex/ coregulator CC/MF terms.

Curation Leads (require curator verification)

- **Candidate references to verify (exact snippets):**

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— "We show that the MAPK of this pathway (Mpk1) and its pseudokinase paralog (Mlp1) use a noncatalytic mechanism to activate transcription of the FKS2 gene." and "Transcriptional activation of FKS2 was dependent on the Swi4/Swi6 (SBF) transcription factor and on an activating signal to Mpk1 but not on protein kinase activity."

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— "In this context, the Mlp1 pseudokinase serves a redundant function with Mpk1."

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— Mpk1–Swi4 complex recruits Swi6 to the nucleus for cell-wall-stress transcription (context for KDX1's shared complex step).

- **Candidate action changes:** Remove / do-not-assert GO:0000307, GO:0004693, GO:0000086 for KDX1 (PAINT over-propagation from CDK clade); downgrade/flag GO:0005524 (IEA-only,

likely impaired); flag GO:0004674/GO:0106310 kinase activity as contradicted by pseudokinase status.

- **Candidate replacement/new terms:** noncatalytic transcription-coregulator role with SBF (Swi4/Swi6); cell-wall-stress transcriptional response / cell wall organization or biogenesis BP; nucleus CC; specific complex/interaction terms for Swi4/Swi6 and Rlm1 (avoid generic "protein binding").
 - **Suggested questions for curator:** Are the disputed IBA terms confirmed to descend from a CDK-clade ancestor (safe to prune)? Is a NOT-kinase-activity qualifier desired? Should ATP binding be retained as IEA or removed pending assay?
 - **Suggested experiments:** nucleotide-binding thermal-shift; kdx1Δ cell-cycle phenotyping; AlphaFold pocket analysis.
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Provenance

Computational analyses executed this run: (1) UniProt REST fetch of P36005 (sequence, features, GO xrefs, comments); (2) UniProt fetch of paralog P32485 (SLT2); (3) regex motif scan comparing catalytic motifs (P-loop, VAIK, HRD, DFG, MAPK T-x-Y) between KDX1 and SLT2. Key computed result: KDX1 lacks GxGxxG, carries HRD→HCD and DFG→NFG substitutions, and has no T-x-Y activation motif, whereas SLT2 retains all. UniProt binding/active-site features for P36005 are ECO:0000255 (PROSITE-ProRule) rule-based, and the disputed CDK/cell-cycle GO terms are IBA:GO_Central; ATP binding is IEA:InterPro. No result was fabricated; where no assay exists (direct ATP binding), this is stated explicitly.
